In Search of a Magic Bottle of Error-Be-Gone



Decoherence

erroso
l

Dave Bacon

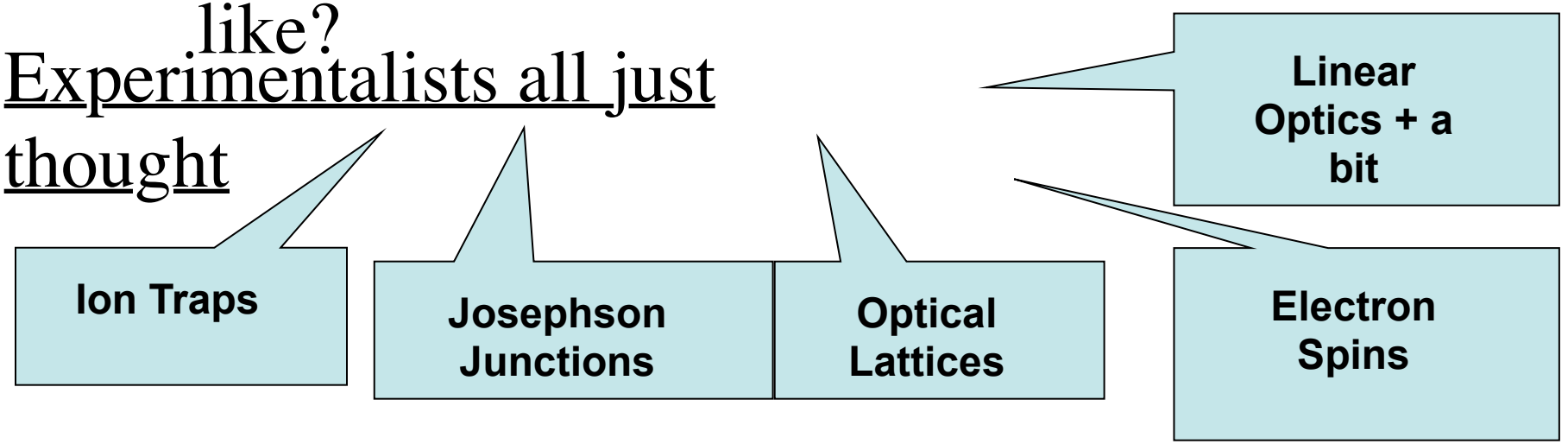
Caltech
Department of Physics
Institute for Quantum
Information

The Future

Visualize 2040:

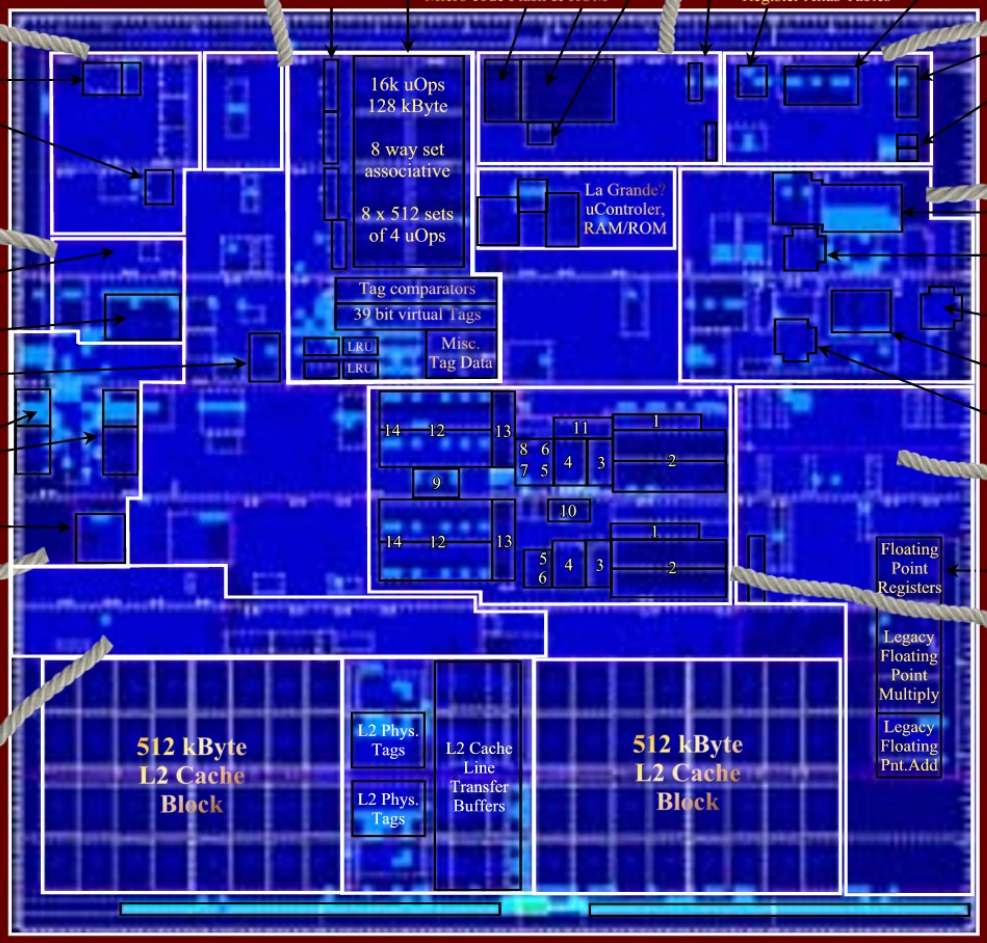
what will a quantum computer look like?

Experimentalists all just thought



Quantum Architecture?

Intel Pentium 5 Prescott



Trace Cache Access, next Address Predict

Trace Cache Branch Prediction Table (BTB), 1024 entries.

Return Stacks (4 x16 entries)

Trace Cache next IP's (4x)

Instruction Trace Cache

Execution Pipeline Start

Buffer Allocation & Register Rename

Instruction Queue (for less critical fields of the uOps)

General Instruction Address Queue & Memory Instruction Address Queue (queues register entries and latency fields of the uOps for scheduling)

Instruction Decoder

Up to 4 decoded uOps/cycle out. (from max. one x86 instr/cycle) Instructions with more than four are handled by Micro Sequencer

Raw Instruction Bytes in

Data TLB, 64 entry fully associative, between threads dual ported (for loads and stores)

Front End Branch Prediction Tables (BTB), shared, 4096 entries in total

Instruction TLB's 128 entry, fully associative for 4k and 4M pages. In: Virtual address [47:12] Out: Physical address [39:12] + 2 page level bits

uOp Schedulers

Parallel (Matrix) Scheduler for the two double pumped ALU's

General Floating Point and Slow Integer Scheduler: (8x8 dependency matrix)

FP Move Scheduler: (8x8 dependency matrix)

Load / Store Linear Address Collision History Table

Load / Store uOp Scheduler: (8x8 dependency matrix)

FP, MMX, SSE1..3

Floating Point, MMX, SSE1..3 Renamed Register File 256 entries of 128 bit.

Instruction Fetch from L2 cache and Branch Prediction

Integer Execution Core

(1) uOp Dispatch unit & Replay Buffer Dispatches up to 6 uOps / cycle

(2) Integer Renamed Register File 256 entries of 32 bit (+ 6 status flags) 12 read ports and six write ports

(3) Databus switch & Bypasses to and from the Integer Register File.

(4) Flags, Write Back

(5) Double Pumped ALU 0

(6) Double Pumped ALU 1

(7) Load Address Generator Unit

(8) Store Address Generator Unit

(9) Load Buffer (96 entries)

(10) Store Buffer (48 entries)

Front Side Bus Interface, 533..800 MHz

(13) Databus multiplexing

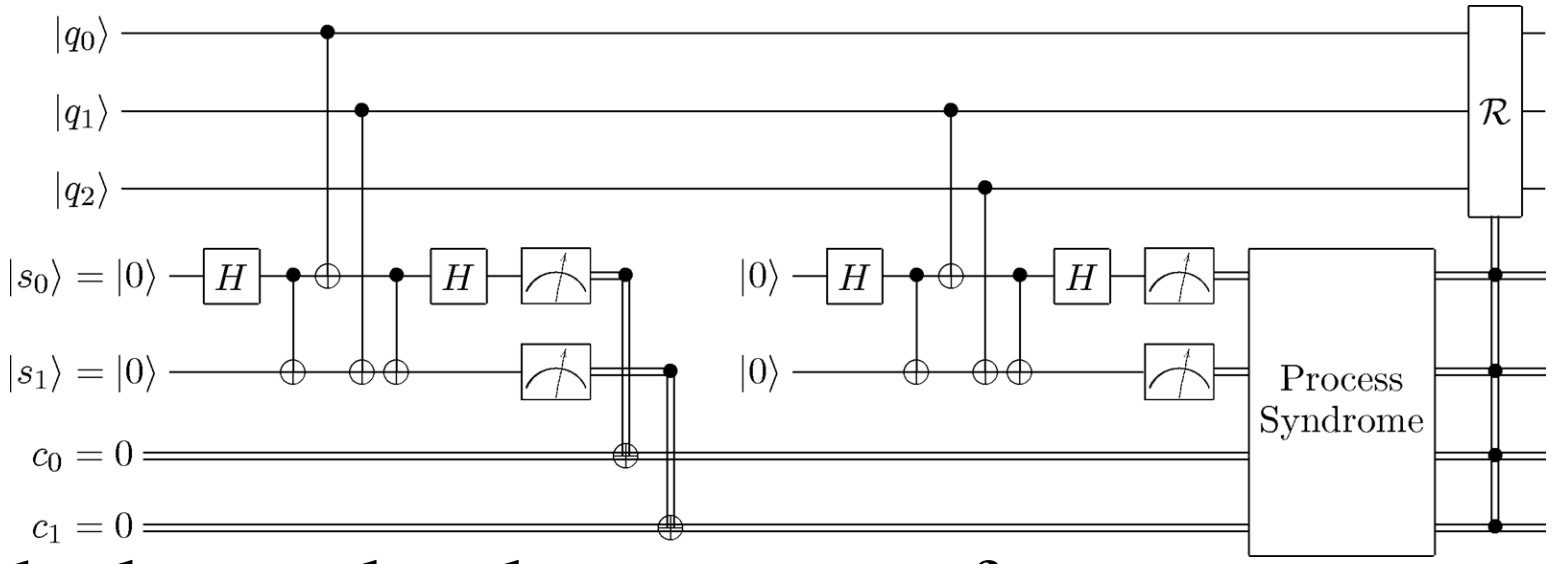
(14) Cache Line Read / Write Transferbuffers and 256 bit wide bus to and from L2 cache

(11) ROB Reorder Buffer 4x64 entries

(12) 16 kByte Level 1 Data cache four way set associative. 1R/1W

April 19, 2003 www.chip-architect.com

Fault Tolerant Quantum Architecture?



At the lowest level we must perform quantum error correction and use fault-tolerant techniques.

Today's talks on quantum error correction will drastically influence what a quantum computer looks like…

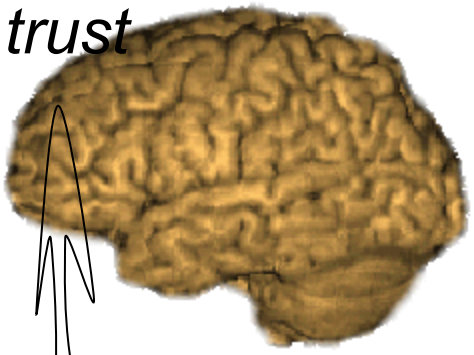
What is the "best" way to do this?

"Best?"

Objection: BEST depends on strengths and weaknesses of particular physical implementations

physical implementation
- gate speeds
- gate accuracies
- gate costs
- forms of decoherence
- decoherence times
- shuttling speeds
- shuttling accuracies
- cooling rates
- calibration errors
- degree of parallelism
- geometric constraints
- fabrication constraints

theory brain trust



quantum architecture specification
- plans suitable for founding qIntel

Two Paths

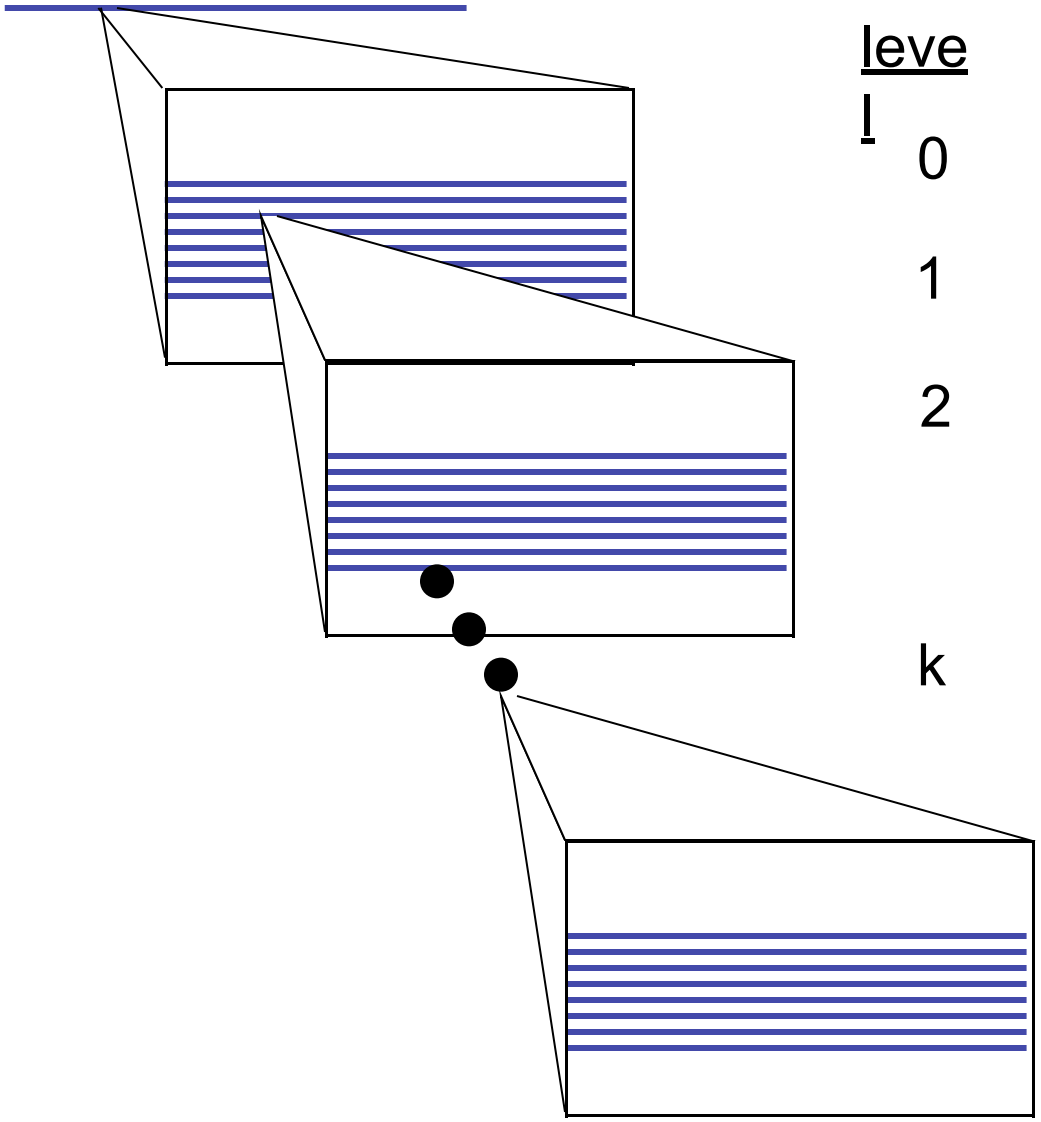
concatenation

local codes



even today TWO *styles* are emerging

Concatenation

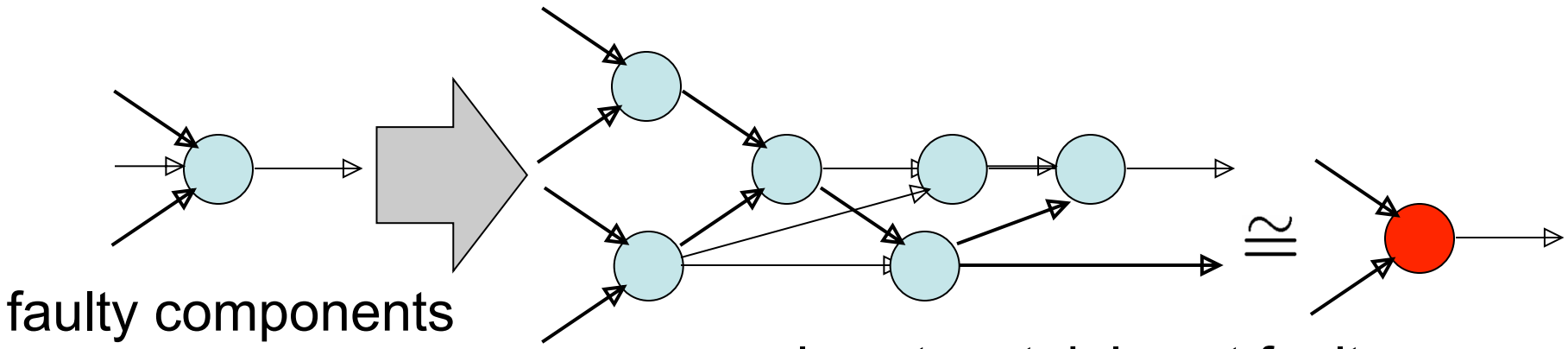


level	qubits	probability of failure
0	1 qubit	p
1	n qubits	cp^2
2	n2 qubits	c(cp2)^2
k	nk qubits	(cp)$^{2^k}$ /c

exponential decrease in # qubits if p<1/c=pthresh

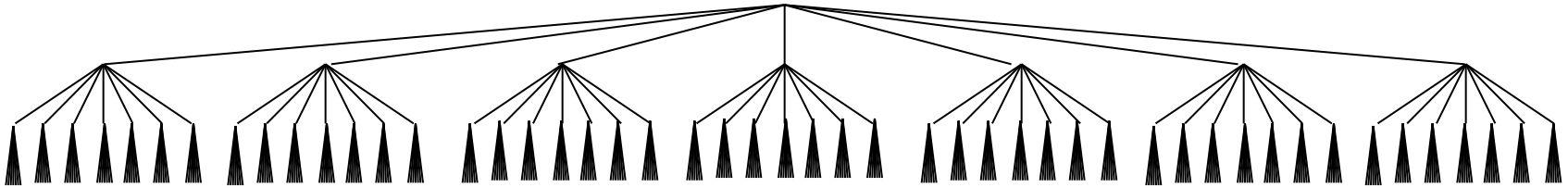
Threshold Theorem

A quantum circuit with
k gates
can be simulated with
error probability e
using
O(k poly(log(k/e))) gates
on hardware whose components fail with probability
p less than some threshold pthresh
under caveats A, B, C, D,…

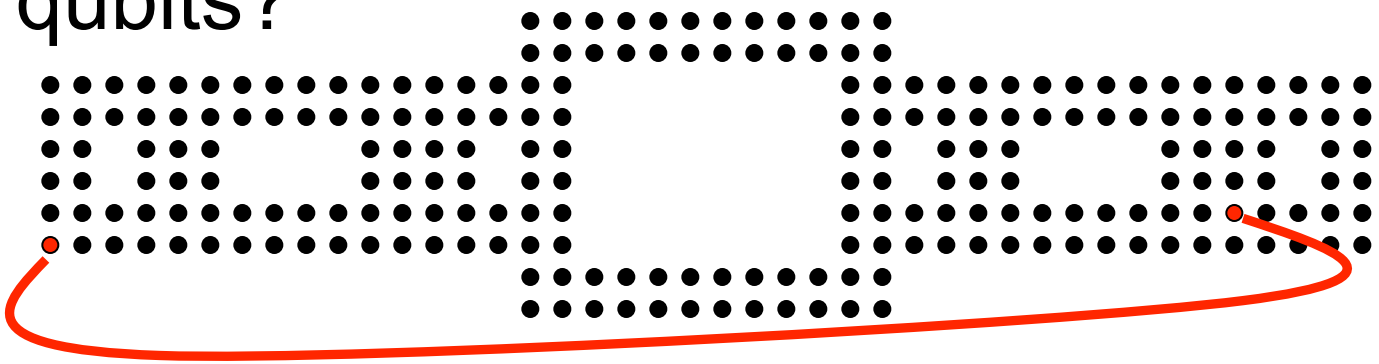


Concatenation and Locality

Concatenation is hierarchical



how to merge with local bare qubits?



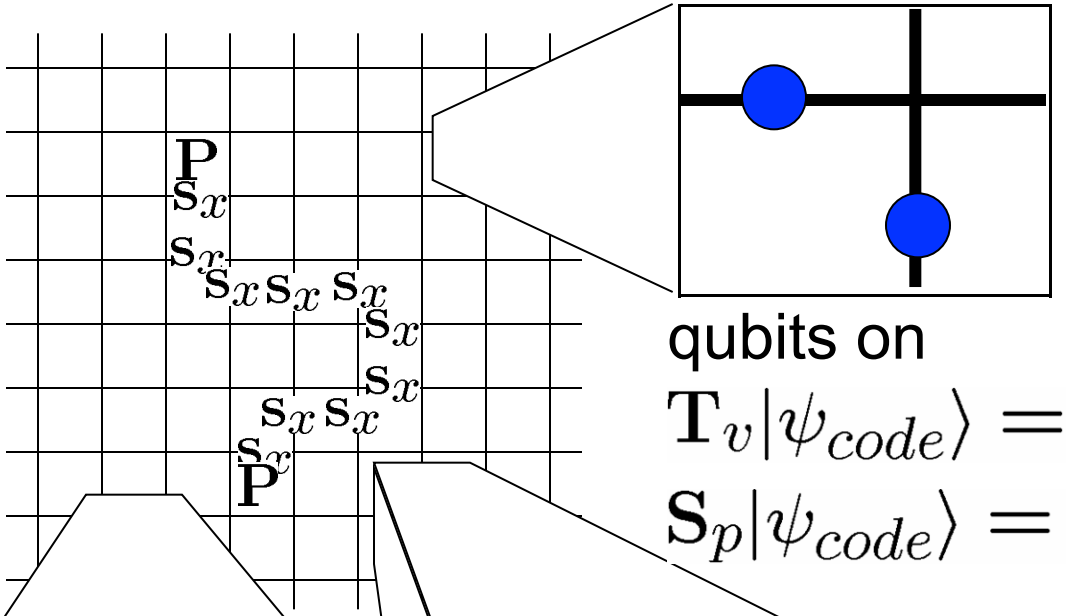
despite: moving or swapping qubits creates error rate proportional to distance moved

THERE IS STILL A THRESHOLD THEOREM

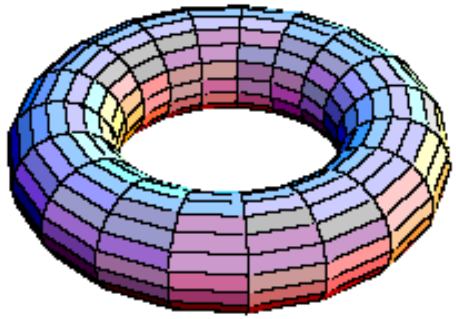
Daniel Gottesman, 1999

Are there non-hierarchical ways to do fault-tolerant quantum computation?

Kitaev's Toric Co



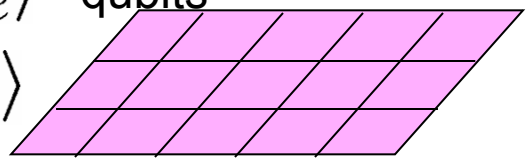
qubits on



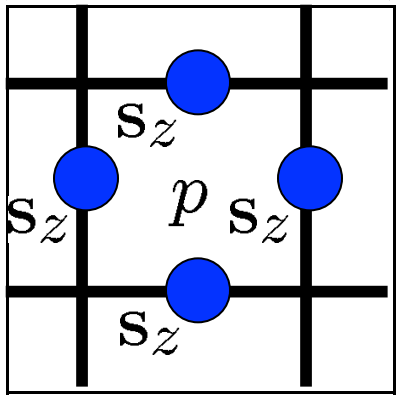
2 encoded qubits

$$\mathbf{T}_v|\psi_{code}\rangle = |\psi_{code}\rangle$$

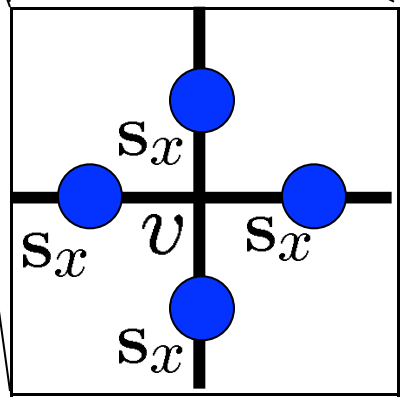
$$\mathbf{S}_p|\psi_{code}\rangle = |\psi_{code}\rangle$$



1 encoded qubit



plaque



vertex operators

$$\mathbf{S}_p = \Pi_{\text{plaque}}\, \mathbf{s}_z \qquad \mathbf{T}_v = \Pi_{\text{vertex}}\, \mathbf{s}_x$$

syndrome measurements involve only four qubit local measurements!

BUT: diagnosing error is not a local process.

Local Codes

Can we find a fully local code?

syndrome

+

diagnosis and correction

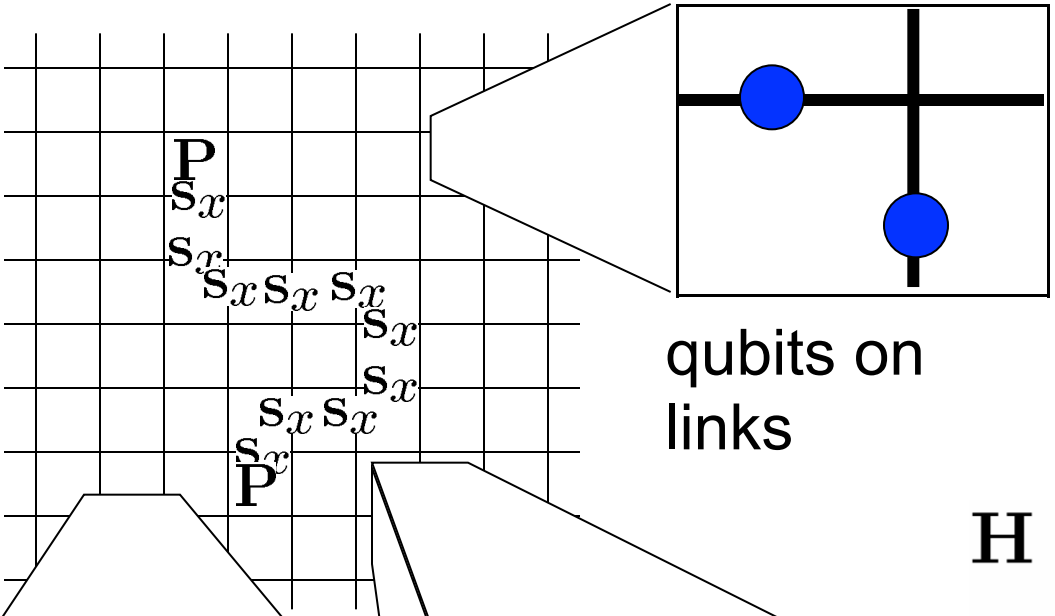
+

fault-tolerant

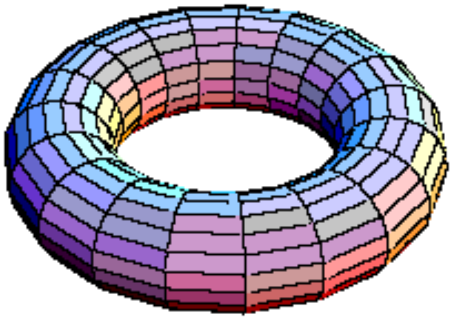
In 4 dimensions there is a fully local code

(sit down silly string theorists)

Physics and Tori



qubits on links



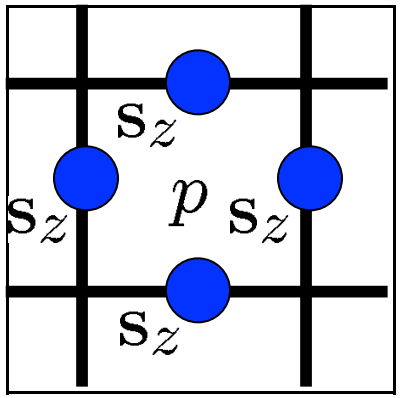
2 encoded qubits

$$\mathbf{H} = -\sum_p \mathbf{S}_p - \sum_v \mathbf{T}_v$$

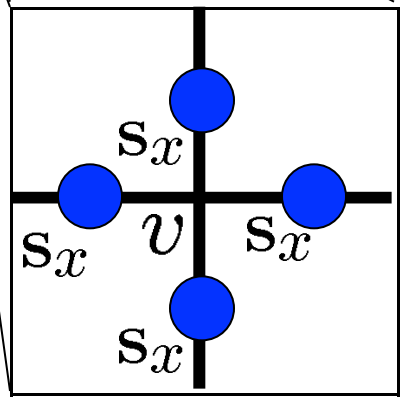
ground state is the toric code!

energy required to excite out
of code: at low temperatures we can freeze out errors

error correction still needed



plaque



vertex operators

$$\mathbf{S}_p = \prod_{\text{plaque}} \mathbf{s}_z \qquad \mathbf{T}_v = \prod_{\text{vertex}} \mathbf{s}_x$$

Rant mode ON

The Physics Guarantee

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation:

there exist systems whose PHYSICS guarantees their ability to enact robust classical computation.

In Practice

Hard Drive



Coding: majority vote of magnetism

Error correction: local energy minimization

Integrated Circuit



Coding: majority vote of current

Error correction: amplification fault-tolerance guaranteed by conducting-insulating phase transition

Rant mode ON

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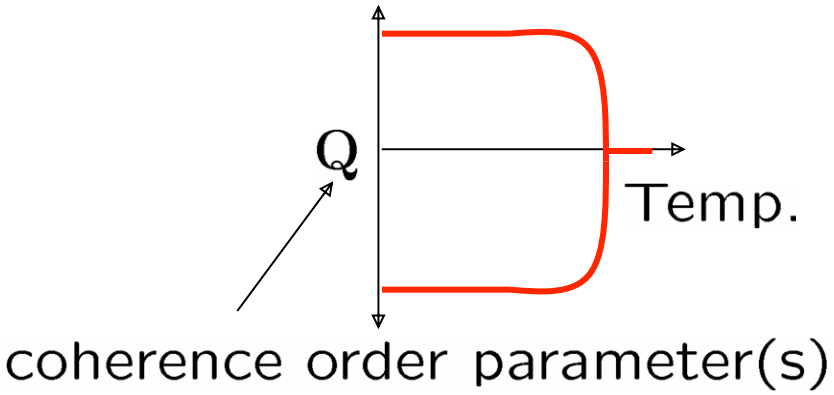
What is the phase of matter corresponding to the quantum computer?

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

Rant mode shutting down

The Quantum Hard Drive?

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.

$Q_1 \rightarrow -Q_1$

macro local operations

2. Accessible Fault-Tolerant Operations

"self-correcting"

$\langle G_1, \ldots, G_k \rangle \cong U(C^{2^n})$

3. Universality

Kitaev's Codes hint that this is possible (in <4D!)

Rant mode OFF

What Will a QC Look Like?





CuBits

concatenation
Engineering?



local
codes
Physics?